

Competition, Heterogeneity, and Inefficiency in Evolving Infrastructure Markets

A Game-Theoretic, Dynamic, and Data-Driven Analysis

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Abstract—We study the efficiency of a competitive gas transportation network in which multiple providers set access prices. This market system cannot be fully described either by a Cournot or a Bertrand framework. Instead, we propose an intermediate modeling approach that captures features of both quantity and price competition. Market outcomes are compared with the definition of a social optimal function, and inefficiencies arising from competition are quantified using the Price of Anarchy. We analyze how equilibrium prices and welfare scale with the number of providers and extend the framework to a heterogeneous setting with the introduction of small and large firms. Ultimately, numerical simulations, complemented by machine-learning-based prediction tools, are used to explore the behavior of the system across a broad parameter space.

Index Terms—Gas transportation networks, game theory, market efficiency, Nash equilibrium, social optimum, price of anarchy, numerical simulations.

I. INTRODUCTION

This work originates from a collaboration with the company *Metanodotti del Friuli*, which suggested a study on the efficiency of gas transportation markets and on the strategies that a medium-sized provider can adopt in a rapidly expanding environment. In the Friuli-Venezia-Giulia region of Italy, the number of active companies in the gas sector increased from approximately 400 to 700 within a few years. Such a rapid entry of new firms raises natural questions regarding competition, pricing strategies, and the overall efficiency of the market.

From a modeling and theoretical perspective, gas transportation markets are characterized by strategic interactions among providers and clients, which are well captured by game-theoretic frameworks. On the one hand, firms compete on prices by offering access to their infrastructure, while capacity constraints, infrastructure costs, and imperfect substitutability between providers play an important role; on the other hand, as *Metanodotti's* report, clients are highly volatile and tend to switch operators based on personal utilities, with an average of about 25% of clients switching every year. As a result, the market cannot be described purely as a Cournot competition nor as a Bertrand competition. In this paper, we therefore adopt an intermediate modeling approach that lies between quantity and price competition and is better suited to represent the structure of gas transportation networks.

The analysis is focused on the providers and is framed within a game-theoretic setting, where each provider chooses

its strategy independently in order to maximize its own profit. Market outcomes resulting from decentralized decision-making are compared with a social utility function. This welfare function takes into account consumer surplus and production costs, and serves as a reference for the most efficient market allocation.

To quantify the inefficiency induced by competition, the welfare function is applied in the derivation of the Price of Anarchy (PoA), defined as the ratio between social welfare at the social optimum and social welfare at the Nash equilibrium,

$$\text{PoA} = \frac{W^{\text{SO}}}{W^{\text{NE}}}. \quad (1)$$

This quantity provides a simple and intuitive measure of how far equilibrium outcomes are from the socially optimal allocation as a result of decentralized strategic behavior.

The framework is also extended to a heterogeneous setting in which firms differ in size, representing small and large providers. Introducing heterogeneity allows us to study how asymmetries in the market affect equilibrium outcomes and inefficiencies.

Finally, given the complexity of the model and the number of relevant parameters, analytical results are complemented by numerical simulations of the birth-death dynamics of providers over time. In addition, machine-learning-based prediction tools are employed to gain further insight into the dynamics of the market.

II. MARKET MODEL

We first consider a market constituted by n identical providers. Each provider i chooses independently a margin (or spread) M_i . We assume that provider i has a client base of value N_i and that competition generates an additional client variation Δ_i driven by relative pricing. We model this variation through a function F ,

$$\Delta_i = F(\bar{M}_{-i} - M_i), \quad \bar{M}_{-i} = \frac{1}{n-1} \sum_{j \neq i} M_j, \quad (2)$$

so that, if provider i sets a lower margin than its competitors, it attracts clients ($\Delta_i > 0$), while a higher margin leads to client losses ($\Delta_i < 0$).

The utility for the providers is defined as

$$u_i(M_i, M_{-i}) = M_i(N_i + \Delta_i) + k \Delta_i. \quad (3)$$

The first term represents the margin times the number of served clients. The second term captures a benefit linked to market expansion / client value. The parameter k measures how much a firm is willing to sacrifice short-run profit to invest in services for the clients: when k is small the focus is mainly on profit, while larger k increases the incentive to attract and retain clients.

A. Homogeneous Social Welfare and Price of Anarchy

We now derive the best response of provider i by differentiating (3) with respect to M_i . Let $x_i = \bar{M}_{-i} - M_i$, so that $\frac{\partial x_i}{\partial M_i} = -1$. Then,

$$\begin{aligned} \frac{\partial u_i}{\partial M_i} &= (N_i + F(x_i)) + M_i F'(x_i) \frac{\partial x_i}{\partial M_i} + k F'(x_i) \frac{\partial x_i}{\partial M_i} \\ &= N_i + F(x_i) - (M_i + k) F'(x_i). \end{aligned} \quad (4)$$

The best response is obtained by setting (4) equal to zero.

We focus on a symmetric equilibrium. Since all companies are of the same type, we impose

$$M_i^* = M^* \quad \forall i, \quad \Rightarrow \quad \bar{M}_{-i} = M^*, \quad x_i = 0. \quad (5)$$

Evaluating the first-order condition at $x_i = 0$ yields

$$0 = N + F(0) - (M^* + k) F'(0), \quad (6)$$

and therefore the symmetric Nash equilibrium margin is

$$M^{\text{NE}} = \frac{N + F(0)}{F'(0)}. \quad (7)$$

a) Choice of F : The function F encodes how clients reallocate across providers as a function of relative margins. Based on previous assumptions it should satisfy: (i) monotonicity (lower margins attract more clients), (ii) saturation (there is a maximum net number of clients that can be gained in a period), and (iii) imperfect substitutability (we are not in a pure Bertrand setting where slightly lower prices immediately capture the whole market). A convenient choice is a bounded, smooth, S-shaped function.

Since Δ_i represents a *variation* in clients, it is natural to impose $F(0) = 0$ (no systematic gain/loss when all margins are equal). A simple choice is a scaled hyperbolic tangent,

$$F_n(x) = \Delta \tanh(b_n x), \quad (8)$$

where $\Delta > 0$ sets the maximum magnitude of net client variation and b_n measures the intensity of competition.

With (8), we have $F_n(0) = 0$ and $F'_n(0) = \Delta b_n$. Plugging into (7) gives

$$M^{\text{NE}} = \frac{N}{\Delta b_n} - k. \quad (9)$$

Finally, we let competition intensity increase with the number of providers but with diminishing sensitivity, and set

$$b_n = \alpha \log(n), \quad (10)$$

which yields the explicit scaling

$$M^{\text{NE}} = \frac{N}{\Delta \alpha \log(n)} - k. \quad (11)$$

b) Price of Anarchy: We can now complete the analysis by studying the Price of Anarchy, as defined in (1). To this end, we introduce a customized social welfare function designed to capture the main factors affecting the efficiency of the gas transportation market.

$$W(M) = -\mu \bar{M} - \lambda \sum_{i=1}^n |\Delta_i| + \eta \sum_{i=1}^n u_i, \quad (12)$$

the first term penalizes high average margins, the second term captures inefficiencies associated with frequent client switching, and the last term promotes the economic sustainability by rewarding high margins.

In order to gain analytical insight, we consider a simplified homogeneous setting and adopt the same approximations introduced in the previous section. In particular, we focus on a regime with negligible switching, so that $\Delta_i = 0$. Under these assumptions, the welfare function reduces to

$$W(M) = -\mu M + \eta n M N. \quad (13)$$

In this simplified setting, the social welfare function is linear in the margin M and therefore does not admit an interior optimum. Therefore, the socially optimal value for the margins lies at the boundary of the admissible strategy space. This prevents the direct computation of a closed-form expression for W^{SO} and, consequently, for the Price of Anarchy in the homogeneous benchmark.

Nevertheless, the characterization of the Nash equilibrium derived above provides useful insight. In particular, the dependence of equilibrium margins on the number of providers allows us to discuss how competition affects market outcomes. We can deduce results that properly aligns with intuition: if the welfare evaluated at the Nash equilibrium improves as the number of providers increases, stronger competition pushes the system toward more efficient outcomes. Conversely, if welfare deteriorates with market fragmentation, increased entry amplifies inefficiencies and leads to higher levels of anarchy.

Importantly, the limitations encountered in the homogeneous setting are a consequence of the simplification of this specific model. When providers are identical, welfare depends only on the common margin and total market activity, and is invariant under reallocations of clients across firms. As a result, inefficiencies related to client switching cannot be fully captured. This observation motivates the introduction of heterogeneity among providers, which breaks the symmetry of the model and restores a meaningful role for client reallocation.

B. Price of Anarchy in a Mixed Market

We now introduce heterogeneity in the static game by considering two types of providers: big firms (B) and small firms (S). The total number of providers is $n = n_B + n_S$. The utility function remains the same as in the homogeneous case, but firms are allowed to differ in their strategic parameter k , which we denote by k_B and k_S . The value k_t measures how much type- t firms value client satisfaction and are willing to sacrifice short-run margin: higher k_t corresponds to a stronger

willingness to invest in the local environment and in services for the client.

In addition, we allow the intensity of competition to differ across types through the parameter α . We set

$$b_t = \alpha_t \log(n), \quad t \in \{B, S\}, \quad (14)$$

where a larger α_t captures a stronger sensitivity of the clients to differences between margins, a higher price responsiveness.

a) Best responses and equilibrium conditions: For a provider of type $t \in \{B, S\}$, we write

$$u_i = M_t(N_t + F_t(x_t)) + k_t F_t(x_t), \quad x_t \equiv \bar{M}_{-t} - M_t \quad (15)$$

where N_t denotes the baseline number of clients served by a type- t provider, and F_t is the switching function. The average margin faced by a type- t provider is

$$\bar{M}_{-t} = \frac{(n_t - 1)M_t + n_{t'}M_{t'}}{n - 1}, \quad t' \neq t \quad (16)$$

Differentiating u_i with respect to M_t yields the first-order condition

$$N_t + F_t(x_t) - (M_t + k_t)F'_t(x_t) = 0, \quad t \in \{B, S\} \quad (17)$$

The mixed-market Nash equilibrium is therefore characterized by the coupled system given by (17) for $t = B$ and $t = S$, together with (16).

b) Local approximation: To obtain a tractable expression, we adopt the same local approximation used in the homogeneous case and assume that equilibrium margins are not too far apart, so that $x_t \simeq 0$. Using the centered specification $F_t(0) = 0$ and $F'_t(0) = \Delta b_t$, the first-order condition (17) reduces to

$$\begin{aligned} N_t - (M_t + k_t)F'_t(0) &\simeq 0, \\ \Rightarrow M_t^{\text{NE}}(n) &\simeq \frac{N_t}{F'_t(0)} - k_t = \frac{N_t}{\Delta \alpha_t \log(n)} - k_t. \end{aligned}$$

Equation (18) implies that a higher k_t leads to lower margin at the equilibrium, consistently with the interpretation of k_t as a stronger preference for client acquisition/retention. Moreover, higher α_t (stronger price responsiveness by the clients) also reduces margins through a more acute switching.

If, for simplicity, we assume $N_B = N_S = N$ and $\alpha_B = \alpha_S = \alpha$, then

$$M_B^{\text{NE}} - M_S^{\text{NE}} \simeq -(k_B - k_S). \quad (18)$$

Therefore, $k_B > k_S$ directly implies $M_B^{\text{NE}} < M_S^{\text{NE}}$, while $k_S > k_B$ implies $M_S^{\text{NE}} < M_B^{\text{NE}}$.

1) Heterogeneous welfare and switching: To estimate the PoA in a mixed market we consider the case $N_B = N_S = N$ and introduce two refinements in the social welfare: (i) a convex penalty on margins, to avoid corner solutions for the social optimum, and (ii) an measure of client switching capturing non-price-driven dynamics observed in practice, as suggested by the company Metanodotti.

We redefine the welfare function as

$$W(\mathbf{M}) = \eta \sum_{i=1}^n u_i - \mu \bar{M} - \lambda \sum_{i=1}^n |\Delta_i| - \frac{\rho}{2} \sum_{i=1}^n M_i^2, \quad (19)$$

where $\bar{M} = \frac{1}{n} \sum_i M_i$ and we set $u_i = M_i(N + \Delta_i)$, retaining the revenue component, and excluding the term $k_t \Delta_i$, which is a private valuation of client acquisition rather than a social component.

We model switching as

$$\Delta_i = F(\bar{M}_{-i} - M_i) + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ i.i.d.} \quad (20)$$

The noise term captures non-price-driven switching: even when margins are similar, a fraction of clients switch provider for non specified reasons.

a) Social optimum under a symmetric approximation: Following Metanodotti's indications, we adopt a symmetric approximation in which margins are close across firms, so that $M_B \simeq M_S \simeq M$ and therefore $F(0) = 0$, yielding $\Delta_i \simeq \varepsilon_i$. Taking expectations in (19) gives

$$\mathbb{E}[W(M)] = \eta n M N - \mu M - \lambda n \mathbb{E}|\varepsilon| - \frac{\rho}{2} n M^2, \quad (21)$$

and using $\mathbb{E}|\varepsilon| = \sigma \sqrt{2/\pi}$ we obtain

$$\mathbb{E}[W(M)] = \eta n M N - \mu M - \lambda n \sigma \sqrt{\frac{2}{\pi}} - \frac{\rho}{2} n M^2. \quad (22)$$

Maximizing (22) yields the socially optimal margin

$$M^{\text{SO}} = \frac{\eta n N - \mu}{\rho n}, \quad (23)$$

(with $M^{\text{SO}} = 0$ if the numerator is negative).

Notably, σ affects the welfare level through a constant loss term, but does not shift the optimizer M^{SO} under this approximation.

b) Nash Equilibrium and PoA: To compute $\mathbb{E}[W^{\text{NE}}]$ consistently with the game-theoretic equilibrium, we evaluate (22) at the Nash equilibrium margin derived in the previous subsection (and at the corresponding mixed-market equilibrium if $M_B^{\text{NE}} \neq M_S^{\text{NE}}$ is considered). The PoA is then given by (1).

2) Analytical Approximations: We derive a tractable approximation by linearizing the switching function around the symmetric point. Denoting $x_i = \bar{M}_{-i} - M_i$, we write

$$S_i \equiv F(x_i) \simeq F'(0) x_i, \quad F'(0) = \Delta \alpha \log(n) \equiv A. \quad (24)$$

Under the local approximation used above, the Nash margins satisfy

$$M_t^{\text{NE}} \simeq \frac{N}{A} - k_t, \quad t \in \{B, S\}. \quad (25)$$

Let $d \equiv M_B - M_S$. We define the relative incentives in switching provider

$$x_B = -\frac{n_S}{n-1}d, \quad x_S = \frac{n_B}{n-1}d, \quad (26)$$

so that

$$S_B \simeq -A \frac{n_S}{n-1}d, \quad S_S \simeq A \frac{n_B}{n-1}d. \quad (27)$$

Using (25), we further obtain $d \simeq -(k_B - k_S) \equiv -\Delta k$.

To connect these expressions to welfare, we divide the churn between intrinsic and excess churn introduced by heterogeneity, focusing on the last one. In the regime where deterministic switching dominates random switching noise ($|S_t| \gg \sigma$ so that $\mathbb{E}[S_t + \varepsilon] \approx |S_t|$), the expected total churn is approximated by

$$\mathbb{E} \left[\sum_i |\Delta_i| \right]_{NE} \simeq n_B |S_B| + n_S |S_S| = 2A \frac{n_B n_S}{n-1} |\Delta k|. \quad (28)$$

Equation (28) highlights three comparative statics: excess churn increases with $\log(n)$, is maximized when the market is mixed ($n_B \simeq n_S$), and grows linearly with the strategic gap $|\Delta k|$. Consequently, the Price of Anarchy reflects a trade-off between the efficiency gains from lower margins as n increases and the welfare losses induced by churn when switching is strongly penalized (large λ).

Overall, the mixed-market analysis shows that heterogeneity transforms competition into a source of excess churn, so that increased entry can either improve or worsen efficiency depending on the relative weight assigned to improved margins and switching costs in social welfare. When the welfare weight on price levels (μ) dominates, increased competition improves efficiency by reducing the margins of the providers. Conversely, when switching costs are heavily penalized (λ large), increased entry can worsen efficiency due to excess churn from the client side.

We can therefore draw some conclusions by this analysis.

Firstly we see that for a given number of providers, excess churn and the corresponding welfare loss are maximized when the market is totally mixed, $n_B \simeq n_S$.

Secondly we understand that for fixed market size and composition, the Price of Anarchy increases with the strategic gap $\Delta k = k_B - k_S$, reflecting a more aggressive client acquisition strategy by one group of providers.

III. PROVIDER ENTRY AND EXIT DYNAMICS

The static analysis developed in the previous sections characterizes market inefficiency as a function of the number and composition of providers. We now extend the framework to a dynamic setting, in order to study how entry and exit of providers shape the long-run evolution of competition and the resulting Price of Anarchy.

We consider a discrete-time model $t = 0, 1, 2, \dots$ in which the market composition evolves through a birth-death process. As before, providers are partitioned into two types: big (B) and small (S). The state of the market at time t is summarized by

$$S_t = (n_B(t), n_S(t)) \in \mathbb{N}^2, \quad (29)$$

where $n(t) = n_B(t) + n_S(t)$ denotes the total number of active providers. This state is sufficient because, as we derived in the static case, equilibrium prices, profits, churn, and entry-exit probabilities depend only on the number and composition of active providers, not on their identities or past histories.

Each period is organized as follows. Given the current state S_t , providers compete in the static pricing game analyzed in Section II.B.1, and a type-symmetric Nash equilibrium is played. Expected profits, churn, welfare, and the Price of Anarchy are then realized as functions of S_t . Subsequently, inefficient providers exit the market with probabilities that depend on their economic performance. Finally, new providers enter the market according to a stochastic entry process, and the state is updated to S_{t+1} .

This structure induces a time-homogeneous Markov chain on $\mathbb{N}^2$. The static model therefore maps the current market composition into equilibrium outcomes,

$$(n_B, n_S) \mapsto (M_B^{NE}, M_S^{NE}, \pi_B, \pi_S, C, \mathbb{E}[W^{SO}], \mathbb{E}[W^{NE}], PoA). \quad (30)$$

which fully determines the transition probabilities of the dynamic process.

A. Numerical Simulation Framework

The long-run behavior of the market cannot be characterized analytically, since entry and exit are stochastic and inefficiency evolves endogenously with market composition. To address this, we study the model through Monte Carlo simulations.

Provider turnover is modeled through a birth-death mechanism designed to capture two dynamics of the markets: (i) bursts of entries driven by external conditions, and (ii) more likely exit for firms with weak profitability and high customer instability.

a) *Death process*: Each incumbent provider of type $t \in \{B, S\}$ exits independently with probability

$$p_t^{\text{death}}(t) = \delta_t \exp(-\gamma_\pi \pi_t(t) + \gamma_c c_t(t)), \quad (31)$$

where $\pi_t(t)$ and $c_t(t)$ are the expected profit and expected churn per provider computed at the Nash equilibrium in the current state $(n_B(t), n_S(t))$. The parameter $\delta_t \in (0, 1)$ captures an intrinsic fragility, and is typically larger for small providers. The sensitivity $\gamma_\pi > 0$ controls how strongly profitability reduces exit risk, while $\gamma_c \geq 0$ controls how strongly customer instability increases exit risk. Conditional on $p_t^{\text{death}}(t)$, the number of exits is

$$D_t(t) \sim \text{Binomial}(n_t(t), p_t^{\text{death}}(t)). \quad (32)$$

b) *Birth process*: Entry is modeled as a Poisson process:

$$B(t) \sim \text{Poisson}(\lambda_B), \quad (33)$$

where λ_B is the average number of new providers entering per period. Conditional on $B(t)$, entrants are assigned to type B with probability q_B :

$$B_B(t) \sim \text{Binomial}(B(t), q_B), \quad B_S(t) = B(t) - B_B(t). \quad (34)$$

c) *State update*: The market state evolves according to

$$\begin{aligned} n_B(t+1) &= n_B(t) - D_B(t) + B_B(t), \\ n_S(t+1) &= n_S(t) - D_S(t) + B_S(t). \end{aligned} \quad (35)$$

which defines a time-homogeneous Markov chain on $\mathbb{N}^2$.

d) *Monte Carlo estimation*: For each parameter configuration, we simulate R independent trajectories over a finite horizon T . We report time series, time averages, and empirical stationary distributions of the main observables: total providers $n(t)$, composition $(n_B(t), n_S(t))$, equilibrium markups, churn, and the Price of Anarchy.

B. Simulation Results

We analyze the dynamic behavior of the market under the birth–death process just introduced. Across all parameter configurations, simulations indicate the emergence of a statistically stable regime, in which the quantities fluctuate around well-defined averages despite persistent stochastic entry and exit.

a) *Market size and stability*: Figure 1 reports the evolution of the total number of active providers $n(t)$. While individual trajectories display increasing dispersion over time, the mean market size remains approximately constant. This indicates that, for the chosen parameters, entry and exit balance each other on average, leading to a quasi-stationary market size.

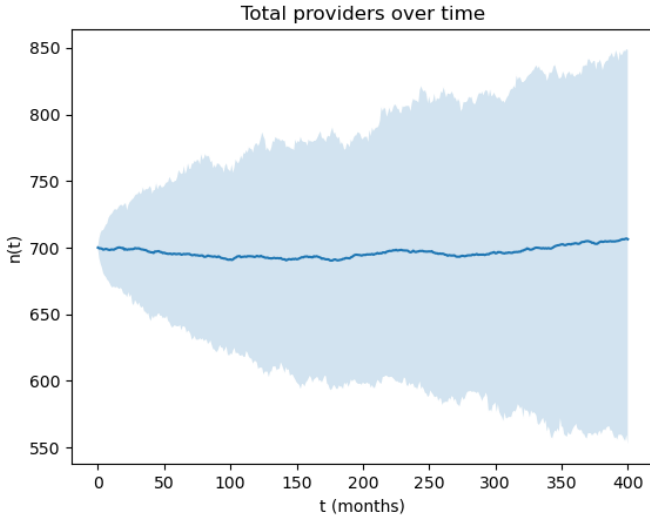


Fig. 1: Total number of active providers over time, indicating a quasi-stationary regime.

b) *Market composition and selection effects*: Despite the stability of total entry, the composition of the market evolves substantially. Figure 2 shows that the share of large providers increases over time and converges to an interior value. This pattern reflects endogenous selection driven by differential exit probabilities: small providers face higher baseline fragility and are more sensitive to churn, while large providers benefit from lower effective markups and more stable customer bases.

Importantly, the market does not converge to a homogeneous structure but stabilizes around a mixed configuration with both types coexisting, validating the relevance of the heterogeneous framework.

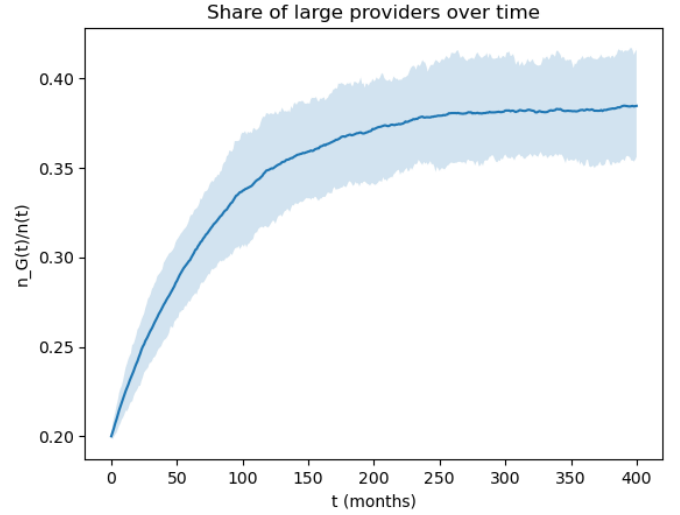


Fig. 2: Share of large providers over time. The market converges to a mixed configuration with a stable, interior fraction of large firms.

c) *Markups and competitive effects*: Figure 3 reports the evolution of equilibrium markups over time. The average Nash markup declines rapidly in the early phases of the simulation, reflecting increased competition following entry, and then stabilizes once the market composition converges. This behavior is consistent with the theoretical prediction $M^{\text{NE}} \sim (\log n)^{-1}$ and confirms that dynamic entry amplifies price competition only up to a point.

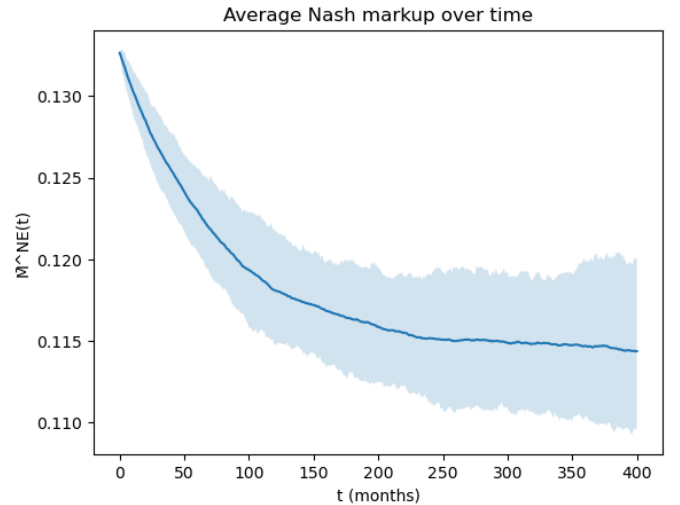


Fig. 3: Average Nash markup and markup dispersion over time.

At the same time, markup dispersion increases and stabilizes at a strictly positive level. This indicates that competition does

not homogenize pricing behavior in the long run; instead, persistent heterogeneity in strategic incentives sustains a stable degree of differentiation in the strategies involving k_B and k_S .

To directly assess the dependence of equilibrium pricing on market size, Figure 4 plots the average Nash markup against $\log n$, to validate qualitatively the expected theoretical scaling. The close alignment confirms that the decline in markups observed over time is primarily driven by the increase in the number of active providers, rather than by transient dynamics.

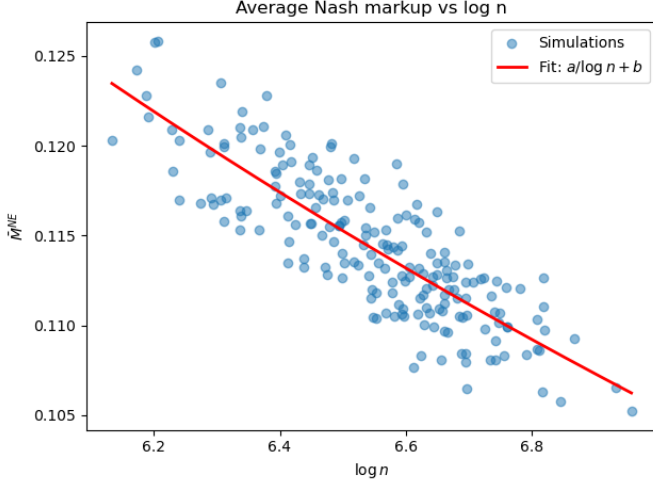


Fig. 4: Average Nash markup as a function of $\log n$, with logarithmic benchmark.

d) Churn and inefficiency: Figure 5 shows that expected churn increases monotonically over time and remains elevated even after the total number of providers stabilizes. This reflects the central mechanism emphasized in the theoretical analysis: inefficiency is driven not only by the number of competitors, but by heterogeneity and strategic asymmetries. As the market converges to a mixed configuration, price-driven switching becomes persistent, generating excess churn relative to the socially optimal benchmark.

To isolate the role of market size, Figure 6 plots expected churn against $\log n$. The approximately linear relationship confirms the analytical prediction that excess churn scales with $\log n$ in heterogeneous markets. As for markups, the logarithmic benchmark is used as a qualitative comparison, highlighting the structural dependence on competition intensity rather than serving as a precise estimator.

e) Price of Anarchy and long-run behavior: The evolution of the Price of Anarchy mirrors the behavior of churn. Figure 7 shows that the PoA increases during the transition phase and stabilizes once the market composition reaches a steady state. Although its absolute level depends on the values assigned to the welfare function and may take negative values due to penalty terms, its dynamic behavior is informative. The long-run inefficiency reflects the studied trade-off between lower margins, which improve efficiency, and rising churn, which worsens it.

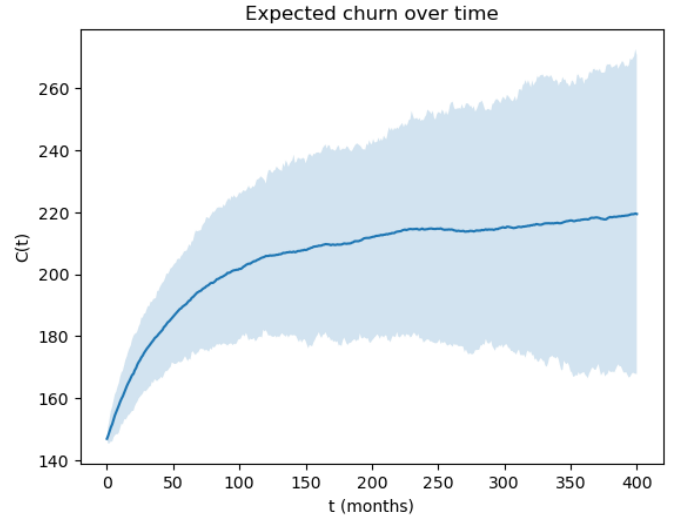


Fig. 5: Expected churn over time.

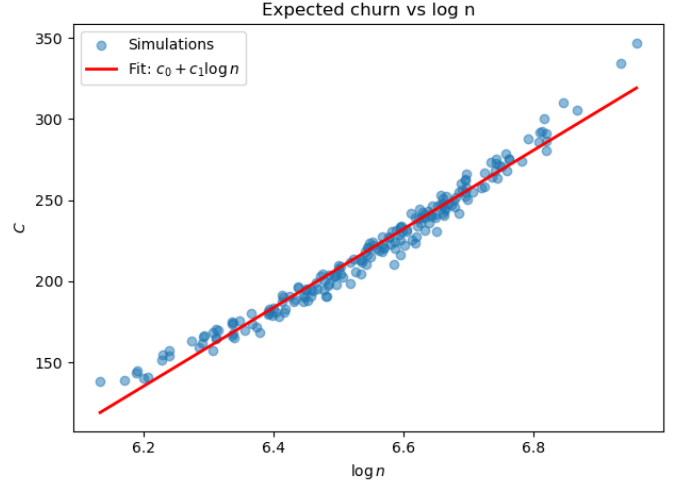


Fig. 6: Expected churn as a function of $\log n$, with logarithmic benchmark.

Stationary distributions further confirm this result. Figure 8 reports the empirical stationary distribution of the PoA, indicating that inefficiency does not diverge over time despite ongoing stochastic entry and exit. Consistent with the analytical predictions, stationary inefficiency increases with the weight assigned to churn in social welfare and with the strategic gap between provider types.

Additional figures reporting diagnostic checks, sensitivity analyses, and alternative visualizations of the dynamics are provided in the appendix.

IV. DATA-DRIVEN ANALYSIS AND MACHINE LEARNING

A. Problem Definition and Data Generation

The analytical results derived in the previous sections characterize equilibrium properties and efficiency losses of the market under different assumptions. From the analysis por-

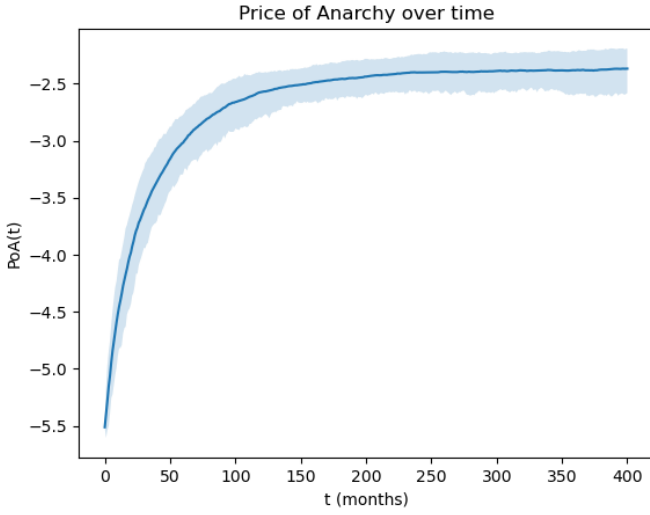


Fig. 7: Price of Anarchy over time. Inefficiency increases during the transition phase and stabilizes once the market reaches a stationary composition.

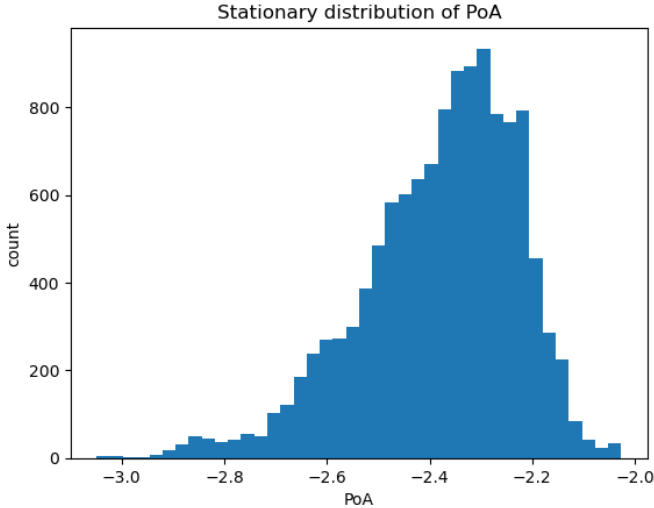


Fig. 8: Stationary distribution of the Price of Anarchy. The distribution is well-defined, indicating persistent but bounded inefficiency in the long run.

trayed we are now interested to understand whether structural properties of the market can be inferred from observable data alone.

In realistic settings, the parameters governing firms' costs and congestion effects are not directly observable. Instead, an external observer typically has access only to a higher layer market quantities, such as inefficiency measures, market concentration, entry and exit dynamics, and customer churn. This motivates the use of data-driven methods to assess the extent to which such observables encode information about the underlying market structure.

Formally, we consider a learning problem in which the inputs consist of statistics extracted from simulated market

dynamics, the targets are the structural parameters of the model. The forward mapping from parameters to observables is implicitly defined by the stochastic dynamics described in Section III. The objective of the machine learning analysis is to investigate whether this mapping can be meaningfully inverted, either to estimate the parameters themselves or to classify qualitatively distinct market regimes.

a) Dataset construction: The learning dataset is generated using the Monte Carlo simulation framework introduced above. For each run, the model is initialized with a randomly sampled parameter vector and evolved over a sufficiently long horizon to reach a stationary regime. From the stationary portion of each trajectory, we extract a set of aggregate observables summarizing both average behavior and temporal fluctuations.

The input data vector includes summary statistics of the Price of Anarchy, average markups, total number of active providers, and customer churn. For each quantity, we compute the empirical mean, standard deviation, and amplitude over the stationary window, resulting in a fixed-dimensional feature space.

The corresponding targets are the structural parameters controlling cost intensity and congestion effects in the model. These parameters also allow for the definition of qualitative market regimes, which will be exploited in subsequent analyses. This procedure yields a synthetic but fully controlled dataset, in which the true data-generating process is known and identifiability can be systematically investigated.

B. Failure of Linear Identification

We begin the data-driven analysis by investigating whether the structural parameters of the model can be inferred from aggregate observables using linear methods.

Specifically, we consider a multivariate linear regression model. The regression is trained on a subset of the simulated dataset and evaluated on a held-out test set, following standard train-validation-test splitting procedures.

Despite the use of enriched feature representations capturing both average behavior and temporal fluctuations, linear regression consistently fails to achieve meaningful predictive performance. On the test set, the coefficient of determination remains close to zero or negative for both parameters (with $R^2 \approx 0$ for θ_{cost} and θ_{cong}), indicating that the linear model performs no better than a constant predictor.

The inadequacy of linear regression therefore motivates the exploration of nonlinear models capable of capturing deeper interactions among observables. At the same time, it raises the possibility that the parameters may not be point-identifiable from stationary aggregate data.

C. Nonlinear Models and Neural Networks

Following the previous results we extend the analysis to nonlinear machine learning models, focusing on neural networks as flexible function approximators.

a) *Model architecture and training:* We consider a fully connected feedforward neural network with two hidden layers and rectified linear unit (ReLU) activations. The input layer corresponds to the aggregate feature vector described in Section IV-A, while the output layer predicts the two structural parameters of interest. The network architecture is intentionally kept shallow and low-dimensional in order to avoid overparameterization and to ensure that any improvement over linear models reflects genuine nonlinear structure rather than overfitting.

The network is trained using mean squared error loss and the Adam optimizer. Standard early stopping based on validation loss is employed to prevent overfitting. All results are evaluated on a held-out test set not used during training or model selection.

b) *Results:* Despite the increased expressive power of the neural network, the nonlinear model fails to achieve meaningful predictive performance on the test set. Predicted parameter values collapse toward their empirical means, and the coefficient of determination remains close to or below zero for both parameters. Scatter plots of predicted versus true values exhibit no discernible alignment along the identity line.

This behavior is indicative of an intrinsic limitation rather than a deficiency of the model. When the neural network is unable to exploit nonlinear interactions to improve upon linear baselines, it suggests that the aggregate observables do not contain sufficient information to uniquely determine the underlying parameters, even under flexible nonlinear representation.

c) *Implications:* The failure of both linear and nonlinear regression models supports the hypothesis that structural parameters are not point-identifiable from stationary aggregate statistics alone. In particular, distinct parameter configurations can generate statistically similar macroscopic market behavior once temporal dynamics are aggregated.

These findings motivate a shift in perspective, leading to the unsupervised analysis presented in the following section, where market regimes are identified through clustering of aggregate observables.

D. Unsupervised Clustering and Emergent Market Regimes

The negative results obtained from both linear and nonlinear supervised learning approaches suggest that instead of attempting to infer latent parameters, we seek to identify recurring patterns of market behavior directly from the observables themselves.

To this end, we adopt an unsupervised learning approach and investigate whether aggregate market statistics naturally cluster into distinct regimes.

a) *Clustering methodology:* We apply k -means clustering to the standardized aggregate feature vectors described in Section IV-A. Clustering is performed on the combined training and validation datasets to maximize statistical robustness, while the test set is reserved for visualization and qualitative assessment.

The number of clusters is selected using the silhouette score, evaluated for values of k ranging from two to six. The

silhouette score varies across this range, reaching a maximum of approximately 0.32 at $k = 4$.

TABLE I: Silhouette score for different numbers of clusters.

Number of clusters k	Silhouette score
2	0.30
3	0.31
4	0.32
5	0.27
6	0.27

This pattern suggests the presence of multiple, moderately well-separated clusters in the data. Accordingly, we focus on a four-cluster partition in the analysis that follows.

b) *Visualization:* For interpretability, we project the feature space onto its first two principal components using principal component analysis (PCA). While the clustering is performed in the full feature space, the two-dimensional projection provides a clear visual representation of the cluster structure. The resulting clusters appear compact and well separated along the leading principal components, confirming the presence of distinct market regimes in the data, characterized by different combinations of competition intensity, churn, and inefficiency.

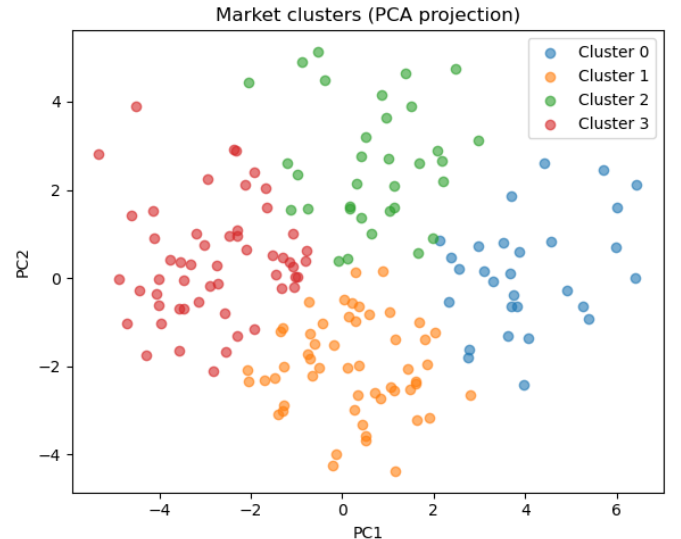


Fig. 9: Projection of the aggregate feature space onto the first two principal components. Colors indicate the four clusters identified by k -means.

c) *Economic interpretation of clusters:* As introduced at the beginning of this section, the precedent analysis aims at capture macro market dynamics by studying aggregate data. We are now able to give an estimate of the four market regimes that emerge from the examination of the cluster-wise averages of the aggregate observables. The identified clusters differ systematically in terms of inefficiency levels, volatility, market size, and customer churn. Some regimes can be associated to relatively small and stable markets with low churn and

limited entry and exit activity, while others capture larger and more turbulent markets characterized by frequent provider turnover and highly volatile inefficiency measures. The other clusters represent competitive yet inefficient configurations, as well as more concentrated markets exhibiting persistently high markups and elevated churn.

Importantly, these regimes are defined purely in terms of observable market behavior and do not rely on knowledge of the underlying structural parameters. This highlights the emergence of robust macroscopic patterns that are directly observable and relevant from both a regulatory and an operational perspective.

d) Relation to structural parameters: The next step is to assess the connection between the emergent clusters and the underlying model parameters, we examine the distribution of cost and congestion parameters within each cluster. We find substantial overlap in parameter distributions across clusters, with only minor differences in their mean values. Most importantly, no cluster is uniquely associated with a specific region of parameter space.

This finding reinforces the conclusions drawn from the supervised learning analysis. Distinct macroscopic market regimes can arise from similar structural parameters, depending on the interaction between competition, entry and exit dynamics, and stochastic fluctuations. Conversely, sharply different observable behaviors doesn't correspond to sharply different parameter values. Overall, these results suggest that regime-based descriptions provide a more informative and robust framework than point estimation of structural parameters from stationary aggregate data.

TABLE II: Qualitative summary of cluster-level averages for key standardized observables.

Cluster	PoA	Churn	Market size n	Avg. markup $\bar{M}$
0	Low	Low	Small	High
1	Moderate	Low	Small	Medium
2	Low	High	Large	Medium
3	High	High	Large	Low

V. CONCLUSIONS

This paper studies competition, inefficiency, and market dynamics in a heterogeneous provider market by combining analytical game-theoretic modeling, stochastic entry-exit simulations, and data-driven learning methods. The objective is duple: to understand how heterogeneity and switching shape efficiency in static and dynamic settings, and to assess whether structural features of the market can be inferred from aggregate observable data.

From a theoretical perspective, the static analysis characterizes Nash equilibria and socially optimal outcomes in markets with heterogeneous providers. A central result is that inefficiency is not driven solely by the level of competition, but by the interaction between market size, heterogeneity in strategic behaviour, and switching inclination. In particular, increased margins can improve efficiency when price distortions

dominate, but may worsen it when churn costs are socially salient. This trade-off is captured analytically by the behavior of the Price of Anarchy as market size grows, and by the emergence of excess switching in mixed markets.

These mechanisms persist in the dynamic setting. Monte Carlo simulations of the stochastic birth-death process reveal rapid convergence toward statistically stable regimes in which aggregate quantities fluctuate around well-defined averages. While the total number of providers remains approximately stable, market composition evolves according to differential exit risk. Large and small providers coexist in the long run, sustaining persistent heterogeneity in markups and churn. Consistent with the static theory, average markups decrease with market size at a rate compatible with a logarithmic law, while churn grows with competition, becoming the dominant source of inefficiency over time.

The dynamic simulations also provide predictive insights at the market level. Over multi-year horizons, the model consistently forecasts stable market size, a non-degenerate mix of provider types, and persistent inefficiency driven by excess switching rather than price distortions. These predictions are robust across parameter configurations and suggest that, under fixed institutional conditions, market outcomes are predictable in distribution even if individual trajectories remain stochastic.

The data-driven analysis complements these findings by addressing the inverse problem of inference from aggregate observables. Both linear regression and nonlinear neural network models fail to recover the underlying structural parameters from stationary market-level statistics.

At the same time, unsupervised clustering reveals that observable market behavior organizes into a small number of robust and interpretable regimes. These regimes differ in terms of inefficiency, volatility, churn, and concentration, despite overlapping parameter distributions. This highlights a key insight of the analysis: while structural parameters are not recoverable from aggregate data, market behavior is nonetheless classifiable in economically meaningful ways.

From an applied perspective, these results carry clear implications for firms and regulators. Strategic decisions and policy interventions should not rely on point estimates of latent parameters inferred from aggregate data. Instead, monitoring market regimes—such as high-churn competitive markets or concentrated inefficient ones—provides a more robust and operationally relevant framework. Firms can use such regime-based diagnostics to anticipate long-run competitive pressure, expected customer turnover, and profitability horizons, as well as the impact of entry policies without requiring full underlying structural model.

Overall, the analysis shows that heterogeneity fundamentally reshapes the relationship between competition and efficiency. While increased competition lowers prices, it can simultaneously amplify socially costly switching. Understanding and managing this trade-off requires moving beyond static efficiency metrics and parameter estimation, toward a dynamic, regime-based view of market behavior.

APPENDIX A

EXPECTED SWITCHING AND WELFARE COMPONENTS

In this appendix we collect the analytical expressions underlying the expected switching terms used in the mixed-market welfare analysis. Consider the switching specification

$$\Delta_i = S + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad (36)$$

where S represents the deterministic component induced by relative margins, and ε_i captures random, non-price-driven switching.

The expected absolute switching is given by

$$g(S, \sigma) \equiv \mathbb{E}[|S + \varepsilon|]. \quad (37)$$

For Gaussian noise, this quantity admits a closed-form expression,

$$g(S, \sigma) = \sigma \sqrt{\frac{2}{\pi}} \exp\left(-\frac{S^2}{2\sigma^2}\right) + S\left(1 - 2\Phi\left(-\frac{S}{\sigma}\right)\right), \quad (38)$$

where $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

The function $g(S, \sigma)$ satisfies $g(0, \sigma) = \sigma\sqrt{2/\pi}$ and is increasing in $|S|$. In the regime $|S| \gg \sigma$, the deterministic component dominates and

$$g(S, \sigma) \simeq |S|. \quad (39)$$

A. Expected Welfare at Nash Equilibrium

In the mixed market, let n_B and n_S denote the number of big and small providers, respectively. Under within-type symmetry, the Nash equilibrium margins are M_B^{NE} and M_S^{NE} , and the expected total welfare at equilibrium is

$$\begin{aligned} \mathbb{E}[W^{\text{NE}}] = & \eta \sum_{t \in \{B, S\}} n_t M_t^{\text{NE}} (N + S_t) \\ & - \mu \bar{M}^{\text{NE}} - \lambda \sum_{t \in \{B, S\}} n_t g(S_t, \sigma) \\ & - \frac{\rho}{2} \sum_{t \in \{B, S\}} n_t (M_t^{\text{NE}})^2. \end{aligned} \quad (40)$$

where

$$\bar{M}^{\text{NE}} = \frac{n_B M_B^{\text{NE}} + n_S M_S^{\text{NE}}}{n}. \quad (41)$$

Using the local approximation developed in the main text,

$$M_t^{\text{NE}} \simeq \frac{N}{\Delta \alpha \log(n)} - k_t, \quad (42)$$

and the expressions for S_B and S_S , equation (40) yields the expected welfare reported in Section IV. The algebraic steps are omitted in the main text for clarity and are collected here for completeness.

APPENDIX B

ADDITIONAL SIMULATION RESULTS

This appendix reports additional numerical results that support the dynamic model analysis and provide further insight into the stochastic properties of the birth-death process. These figures complement the main text but are not essential for understanding the core mechanisms driving inefficiency.

APPENDIX

A. Phase-Space Dynamics

Figure 10 reports the trajectories of the market state $(n_B(t), n_S(t))$ in phase space across multiple Monte Carlo runs. Each trajectory corresponds to a different realization of the stochastic entry–exit process.

Despite substantial fluctuations, trajectories remain confined to a compact region of the state space and do not diverge toward explosive growth or market collapse. This visual evidence supports the existence of a stationary distribution over market compositions and justifies the focus on long-run averages and stationary statistics in the main analysis.

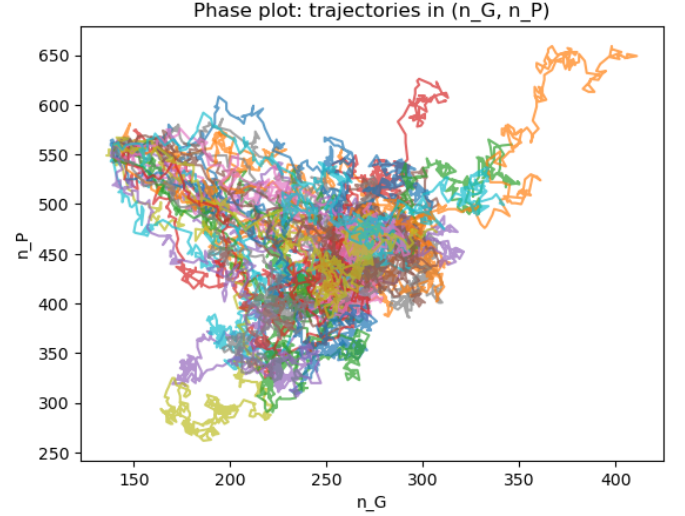


Fig. 10: Phase-space trajectories of the birth–death dynamics in the (n_B, n_S) plane across multiple Monte Carlo runs.

B. Provider Lifetimes and Exit Risk

Figure 11 reports a result for expected provider lifetimes by type, computed from realized exit frequencies over time. Large providers exhibit significantly longer expected lifetimes than small providers, reflecting lower baseline fragility and higher resilience to churn. Small providers, by contrast, face consistently shorter lifetimes, which contributes to the endogenous selection dynamics discussed in the main text.

C. Market Robustness

Figure 12 reports the empirical probability that the total number of providers falls below a critical threshold. Across all simulations, this probability remains low, indicating that the market is robust to stochastic shocks and does not exhibit systemic collapse under the baseline parameterization. This finding confirms that the birth-death dynamics generate persistent fluctuations around a stable regime rather than unstable cycles that might lead to collapse.

This appendix provides additional diagnostics for the non-linear machine learning analysis presented in Section IV-C. Figure 13 reports scatter plots of predicted versus true parameter values for the neural network evaluated on the held-out test

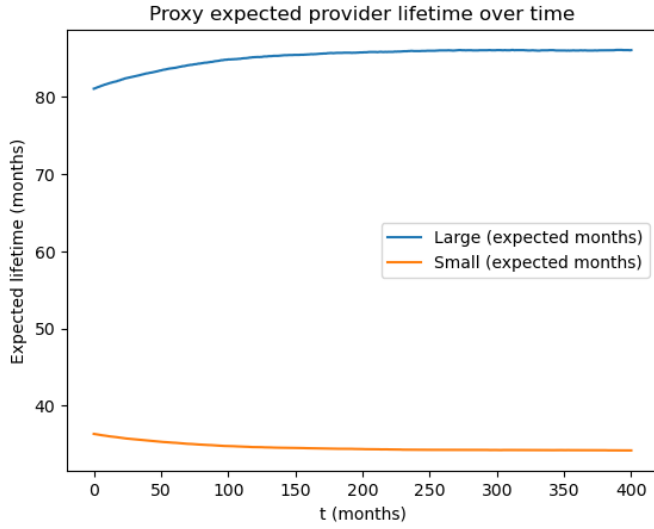


Fig. 11: Expected provider lifetimes by type.

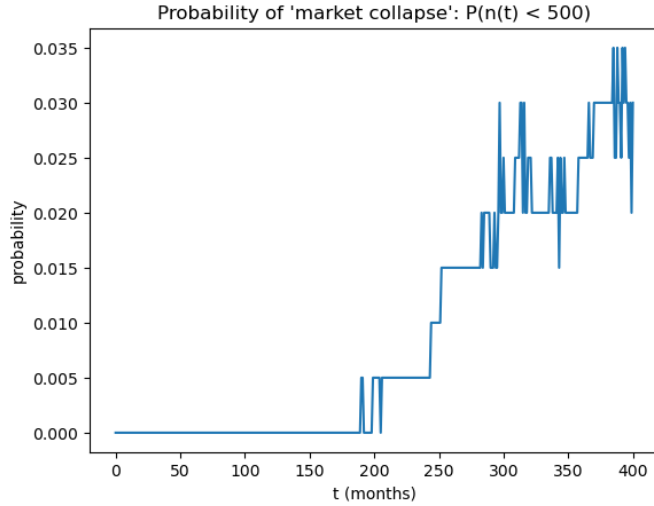


Fig. 12: Empirical probability of market collapse, defined as the event that the total number of providers falls below a critical threshold.

set. Each point corresponds to a simulated market realization, while the dashed line represents perfect prediction.

For both parameters, predictions collapse toward their empirical means and exhibit no systematic alignment along the identity line. Consistently, the coefficient of determination on the test set is negative, with

$$R_{\theta_{\text{cost}}}^2 = -0.16, \quad R_{\theta_{\text{cong}}}^2 = -0.26,$$

indicating that the neural network performs worse than a constant mean predictor.

These results confirm that the failure observed in the main text is not due to insufficient model capacity or optimization, but reflects an intrinsic limitation of stationary aggregate observables, which do not contain enough information to uniquely identify the underlying structural parameters.

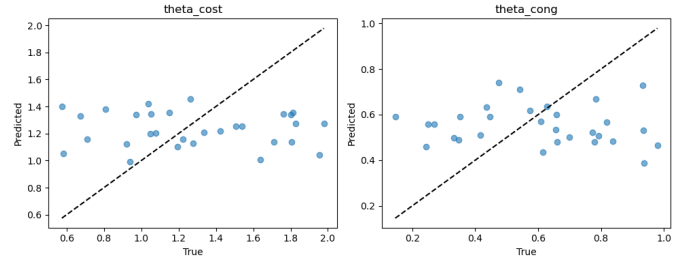


Fig. 13: Neural network predictions versus true parameter values on the test set.

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